Churn Analysis and Customer Intelligence

In [12]:
```
# Churn Analysis and Customer Intelligence
```

In [13]:
```
# import libraries
import numpy as np
import pandas as pd
import matplotlib.pyplot as plt
import seaborn as sns
import sqlite3
```

1. Import database / data

In [14]:
```
# Data Import: db file to pandas, storing each table to a separate df

# Connect to SQLite database
conn = sqlite3.connect('customer_churn.db')

# sql query to Get all table names
sql_query = """
        SELECT name
        FROM sqlite_master
        WHERE type='table';
"""

# read sql query in pandas
tables = pd.read_sql(sql_query, conn)

# create dataframe for each table
for table_name in tables['name']:
    df = pd.read_sql(f"SELECT * FROM {table_name}", conn) # Read table into data
    globals()[f"df_{table_name}"] = df                    # Create dynamic dataf
    print(f"Created dataframe: df_{table_name}")

# Close connection
conn.close()
```

```
Created dataframe: df_db_customer
Created dataframe: df_db_subscription
Created dataframe: df_db_support
```

In [15]:
```
# Print table names and column names
conn = sqlite3.connect('customer_churn.db')

for table_name in tables['name']:
    print(f"\nTable Name: {table_name}")
    # Get column information
    columns_query = f"PRAGMA table_info({table_name});"
    columns = pd.read_sql(columns_query, conn)
    print("Columns:")
    print(columns['name'].tolist())

# Close connection
conn.close()
```

```
Table Name: db_customer
Columns:
['customerid', 'name', 'country', 'state', 'gender', 'dob', 'interests', 'pincod
e']

Table Name: db_subscription
Columns:
['customerid', 'subscription_start_date', 'subscription_type', 'renewal_date', 'p
lan_type', 'contract_type', 'cancellation_date', 'cancellation_reason', 'monthly_
charges', 'cltv', 'churn_score']

Table Name: db_support
Columns:
['customerid', 'complaint_date', 'escalations', 'csat_score', 'col_1', 'comment']
```

In [16]:
```
# PRAGMA is a special command in SQLite used to:
# inspect db information, control db settings, retrieve metadata about tables
```

2. Data cleaning

In [17]:
```
# df_db_customer.head() # customer table
df_db_customer.tail()
```

Out[17]:

	customerid	name	country	state	gender	dob	interests	pincode
16	0020-JDNXP	rikim	India	Meghalaya	Female	1994-08-19 00:00:00	NaN	None
17	0021-IKXGC	vishakha	India	Rajasthan	Female	2000-09-02 00:00:00	NaN	None
18	0022-TCJCI	raghvendra	India	Telangana	Male	1983-12-30 00:00:00	NaN	None
19	0023-HGHWL	rishabh	India	Uttar Pradesh	Men	1991-05-14 00:00:00	NaN	None
20	0023-UYUPN	sudevi	India	Maharashtra	Women	1977-10-06 00:00:00	NaN	None

In [18]:
```
df_db_customer.info()
```

```
<class 'pandas.DataFrame'>
RangeIndex: 21 entries, 0 to 20
Data columns (total 8 columns):
 #   Column      Non-Null Count  Dtype
---  ------      --------------  -----
 0   customerid  21 non-null     str
 1   name        21 non-null     str
 2   country     18 non-null     str
 3   state       21 non-null     str
 4   gender      21 non-null     str
 5   dob         21 non-null     str
 6   interests   4 non-null      str
 7   pincode     0 non-null      object
dtypes: object(1), str(7)
memory usage: 1.4+ KB
```

In [19]:
```
# a. rename col - name to customer_name
# b. drop columns - interest and pincode
# c. change data type - dob
# d. data standardization - gender
# e. fix missing values (using existing data) - country
```

a. Rename Col

In [20]:
```
# a. rename col - name to customer_name

df_db_customer.rename(columns = {'name' : 'customer_name'}, inplace= True)
```

b. Drop columns

In [21]:
```
# b. drop columns - interest and pincode

# df_db_customer.drop(df_db_customer.columns[-2:], axis=1)
# df_db_customer.drop(df_db_customer.columns[6:], axis=1)

df_db_customer.drop(columns=['interests', 'pincode'], inplace=True)  # using col
```

In [22]:
```
df_db_customer.info()
```

```
<class 'pandas.DataFrame'>
RangeIndex: 21 entries, 0 to 20
Data columns (total 6 columns):
 #   Column         Non-Null Count  Dtype
---  ------         --------------  -----
 0   customerid     21 non-null     str
 1   customer_name  21 non-null     str
 2   country        18 non-null     str
 3   state          21 non-null     str
 4   gender         21 non-null     str
 5   dob            21 non-null     str
dtypes: str(6)
memory usage: 1.1 KB
```

c. Change data type

In [23]:
```
# c. change data type - dob

df_db_customer['dob'] = pd.to_datetime(df_db_customer['dob'])
```

d. Data standardization¶

In [24]:
```
# d. data standardization - gender

# df_db_customer['gender'].unique()
df_db_customer['gender'] = df_db_customer['gender'].replace({'Men' : 'Male', 'Wo
```

In [25]:
```
df_db_customer['gender'].unique()
```

Out[25]:
```
<StringArray>
['Male', 'Female']
Length: 2, dtype: str
```

e. Fix missing values

In [26]:
```
# e. fix missing values - country

# df_db_customer['country'].isna().sum()
df_db_customer[df_db_customer['country'].isna()]
```

Out[26]:

	customerid	customer_name	country	state	gender	dob
5	0013-MHZWF	durga	NaN	Delhi	Female	1988-12-10
8	0015-UOCOJ	maya	NaN	Kathmandu	Female	1985-07-07
12	0018-NYROU	chitra	NaN	Telangana	Female	2004-12-01

In [27]:
```
# country and state - unique value pair

# Creating state → country map from non-null rows
state_country_mapping = df_db_customer.dropna(subset=['country']).set_index('sta

# Fill the missing country using State
df_db_customer['country'] = df_db_customer['country'].fillna(df_db_customer['sta
```

In [28]:
```
df_db_customer[df_db_customer['country'].isna()] # no null value in country col
```

Out[28]:

	customerid	customer_name	country	state	gender	dob

In [ ]:
```

```

In [29]:
```
df_db_subscription.head() # subcription table
```

Out[29]:

	customerid	subscription_start_date	subscription_type	renewal_date	plan_type	cont
0	0002-ORFBO	2021-03-15	Refferal	2025-03-15	Standard	
1	0003-MKNFE	2020-08-01	Paid	2024-08-01	Premium	
2	0004-TLHLJ	2022-11-20	Organic	2025-11-20	Basic	
3	0011-IGKFF	2019-05-10	Paid	2025-05-10	Premium	
4	0013-EXCHZ	2023-01-05	Refferal	2024-01-05	Standard	

In [30]:
```
df_db_subscription.info()
```

```
<class 'pandas.DataFrame'>
RangeIndex: 21 entries, 0 to 20
Data columns (total 11 columns):
 #   Column                  Non-Null Count  Dtype
---  ------                  --------------  -----
 0   customerid              21 non-null     str
 1   subscription_start_date 21 non-null     str
 2   subscription_type       21 non-null     str
 3   renewal_date            21 non-null     str
 4   plan_type               21 non-null     str
 5   contract_type           21 non-null     str
 6   cancellation_date       6 non-null      str
 7   cancellation_reason     6 non-null      str
 8   monthly_charges         21 non-null     float64
 9   cltv                    21 non-null     int64
 10  churn_score             21 non-null     int64
dtypes: float64(1), int64(2), str(8)
memory usage: 1.9 KB
```

In [31]:
```
# change data type to date - subscription_start_date , renewal_date, cancellatio
date_col = ['subscription_start_date', 'renewal_date', 'cancellation_date']

df_db_subscription[date_col] = df_db_subscription[date_col].apply(pd.to_datetime
df_db_subscription.info()
```

```
<class 'pandas.DataFrame'>
RangeIndex: 21 entries, 0 to 20
Data columns (total 11 columns):
 #   Column                   Non-Null Count  Dtype
---  ------                   --------------  -----
 0   customerid               21 non-null     str
 1   subscription_start_date  21 non-null     datetime64[us]
 2   subscription_type        21 non-null     str
 3   renewal_date             21 non-null     datetime64[us]
 4   plan_type                21 non-null     str
 5   contract_type            21 non-null     str
 6   cancellation_date        6 non-null      datetime64[us]
 7   cancellation_reason      6 non-null      str
 8   monthly_charges          21 non-null     float64
 9   cltv                     21 non-null     int64
 10  churn_score              21 non-null     int64
dtypes: datetime64[us](3), float64(1), int64(2), str(5)
memory usage: 1.9 KB
```

In [ ]:

In [32]:
```
df_db_support.head() # support table
```

Out[32]:

	customerid	complaint_date	escalations	csat_score	col_1	comment
0	0003-MKNFE	2024-08-28 00:00:00	N	60	None	service issue
1	0003-MKNFE	2024-08-28 00:00:00	Y	10	None	demaned refund
2	0013-EXCHZ	2024-01-20 00:00:00	Y	20	None	NaN
3	0013-MHZWF	2025-03-18 00:00:00	N	90	None	guidance to renew
4	0013-SMEOE	2024-11-01 00:00:00	N	30	None	NaN

In [33]:
```
df_db_support.info()
```

```
<class 'pandas.DataFrame'>
RangeIndex: 9 entries, 0 to 8
Data columns (total 6 columns):
 #   Column          Non-Null Count  Dtype
---  ------          --------------  -----
 0   customerid      9 non-null      str
 1   complaint_date  9 non-null      str
 2   escalations     9 non-null      str
 3   csat_score      9 non-null      int64
 4   col_1           0 non-null      object
 5   comment         4 non-null      str
dtypes: int64(1), object(1), str(4)
memory usage: 564.0+ bytes
```

In [34]:
```
# drop columns
df_db_support.drop(columns=['col_1', 'comment'], inplace=True)
```

In [35]:
```
# change data type to date - complaint_date

df_db_support['complaint_date'] = pd.to_datetime(df_db_support['complaint_date']
df_db_support.info()
```

```
<class 'pandas.DataFrame'>
RangeIndex: 9 entries, 0 to 8
Data columns (total 4 columns):
 #   Column          Non-Null Count  Dtype
---  ------          --------------  -----
 0   customerid      9 non-null      str
 1   complaint_date  9 non-null      datetime64[us]
 2   escalations     9 non-null      str
 3   csat_score      9 non-null      int64
dtypes: datetime64[us](1), int64(1), str(2)
memory usage: 420.0 bytes
```

In [ ]:

3. Feature Engineering & Data Analysis

Create a new col

In [36]:
```
# create a new col using existing col - churn flag

# Customer is churned if cancellation_date is not null
df_db_subscription['churn_flag'] = np.where(df_db_subscription['cancellation_dat
df_db_subscription.head()
```

Out[36]:

	customerid	subscription_start_date	subscription_type	renewal_date	plan_type	cont
0	0002-ORFBO	2021-03-15	Refferal	2025-03-15	Standard	
1	0003-MKNFE	2020-08-01	Paid	2024-08-01	Premium	
2	0004-TLHLJ	2022-11-20	Organic	2025-11-20	Basic	
3	0011-IGKFF	2019-05-10	Paid	2025-05-10	Premium	
4	0013-EXCHZ	2023-01-05	Refferal	2024-01-05	Standard	

In [ ]:

Merge Dataframes - JOINs

In [37]:
```
# first fix support table duplicates then merge
# Note: while merging df's always check the shape before and after
df = (df_db_subscription
```

```
            .merge(df_db_customer, on = 'customerid', how= 'left')
            .merge(df_db_support, on = 'customerid', how= 'left') )
```

In [38]:
```
df_db_subscription.shape
```

Out[38]: (21, 12)

In [39]:
```
df.shape
```

Out[39]: (23, 20)

In [40]:
```
print('df_db_subscription unique value:', df_db_subscription['customerid'].nuniq
print('df_db_customer unique value:', df_db_customer['customerid'].nunique())
print('df_db_support unique value:', df_db_support['customerid'].nunique())
print('df_db_support all value:', df_db_support['customerid'].size)
```

```
df_db_subscription unique value: 21
df_db_customer unique value: 21
df_db_support unique value: 7
df_db_support all value: 9
```

In [41]:
```
df_db_support
```

Out[41]:

	customerid	complaint_date	escalations	csat_score
0	0003-MKNFE	2024-08-28	N	60
1	0003-MKNFE	2024-08-28	Y	10
2	0013-EXCHZ	2024-01-20	Y	20
3	0013-MHZWF	2025-03-18	N	90
4	0013-SMEOE	2024-11-01	N	30
5	0017-IUDMW	2024-04-10	Y	25
6	0019-EFAEP	2024-09-27	Y	30
7	0022-TCJCI	2024-09-13	Y	10
8	0022-TCJCI	2024-09-14	N	90

In [42]:
```
df_db_support['complaint_count'] = df_db_support.groupby('customerid')['customer
```

In [43]:
```
df_db_support = df_db_support.sort_values('complaint_date').drop_duplicates('cus
```

In [44]:
```
df_db_support['customerid'].size
```

Out[44]: 7

In [45]:
```
# merge df
df = (df_db_subscription
            .merge(df_db_customer, on = 'customerid', how= 'left')
            .merge(df_db_support, on = 'customerid', how= 'left') )
```

In [46]:
```
df
```

Out[46]:

	customerid	subscription_start_date	subscription_type	renewal_date	plan_type	cor
0	0002-ORFBO	2021-03-15	Refferal	2025-03-15	Standard	
1	0003-MKNFE	2020-08-01	Paid	2024-08-01	Premium	
2	0004-TLHLJ	2022-11-20	Organic	2025-11-20	Basic	
3	0011-IGKFF	2019-05-10	Paid	2025-05-10	Premium	
4	0013-EXCHZ	2023-01-05	Refferal	2024-01-05	Standard	
5	0013-MHZWF	2022-06-18	Paid	2025-06-18	Standard	
6	0013-SMEOE	2021-09-30	Refferal	2024-09-30	Basic	
7	0014-BMAQU	2020-02-14	Organic	2025-02-14	Premium	
8	0015-UOCOJ	2023-07-22	Organic	2024-07-22	Standard	
9	0016-QLJIS	2022-04-03	Organic	2025-04-03	Basic	
10	0017-DINOC	2021-12-01	Organic	2025-12-01	Premium	
11	0017-IUDMW	2023-03-17	Refferal	2024-03-17	Standard	
12	0018-NYROU	2020-10-09	Organic	2025-10-09	Standard	
13	0019-EFAEP	2022-08-25	Refferal	2024-08-25	Basic	
14	0019-GFNTW	2019-11-11	Paid	2025-11-11	Premium	
15	0020-INWCK	2023-05-06	Organic	2025-05-06	Standard	
16	0020-JDNXP	2022-01-19	Organic	2024-01-19	Premium	
17	0021-IKXGC	2021-07-07	Paid	2025-07-07	Standard	
18	0022-TCJCI	2023-09-14	Refferal	2024-09-14	Basic	
19	0023-HGHWL	2020-06-23	Organic	2025-06-23	Premium	
20	0023-UYUPN	2022-12-31	Paid	2025-12-31	Standard	

21 rows × 21 columns

In [47]:
```
df.shape
```

Out[47]: (21, 21)

Export dataframe

In [48]:
```
# export dataframe to csv file
df.to_csv('exported_churn_data.csv', index=False)
```

In [ ]:
```

```

Data Analysis

In [49]:
```
# 1. Churn Rate
```

In [50]:
```
df.columns
```

Out[50]:
```
Index(['customerid', 'subscription_start_date', 'subscription_type',
       'renewal_date', 'plan_type', 'contract_type', 'cancellation_date',
       'cancellation_reason', 'monthly_charges', 'cltv', 'churn_score',
       'churn_flag', 'customer_name', 'country', 'state', 'gender', 'dob',
       'complaint_date', 'escalations', 'csat_score', 'complaint_count'],
      dtype='str')
```

In [51]:
```
churn_rate = df['churn_flag'].mean()*100
print("Churn Rate = ", round(churn_rate,2), "%")
```

```
Churn Rate =  28.57 %
```

In [52]:
```
# 2.Retenation Rate
retention_rate = 100 - churn_rate
print("Retention Rate = ", round(retention_rate,2), "%")
```

```
Retention Rate =  71.43 %
```

In [53]:
```
# 3. Churn by  plan type
churn_by_plan = (df.groupby('plan_type')['churn_flag'].mean().mul(100).round(2).
print(churn_by_plan)
```

```
  plan_type  Churn_rate_pct
0     Basic           60.00
1   Premium           14.29
2  Standard           22.22
```

4 a.Churn by state + sum(revenue) & count of users

In [136…
```
# Churn by state + sum(revenue) & count of users
```

```
churn_by_state = df.groupby(['state', 'churn_flag']).agg(
total_revenue=('monthly_charges', 'sum'),
user_count=('customerid', 'count')
).reset_index()
print(churn_by_state)
```

	state	churn_flag	total_revenue	user_count
0	Delhi	0	38.97	3
1	Delhi	1	13.99	1
2	Karnataka	1	20.98	2
3	Kathmandu	0	20.98	2
4	Maharashtra	0	50.97	3
5	Meghalaya	0	20.99	1
6	Meghalaya	1	21.98	2
7	Nagaland	0	22.99	1
8	Rajasthan	0	36.98	2
9	Telangana	0	13.99	1
10	Telangana	1	16.99	1
11	Uttar Pradesh	0	115.98	2

4 b. Churn by subscription type + sum(revenue) & count of users

In [137…

```
churn_by_subscription = df.groupby(['subscription_type','churn_flag']).agg(
    total_revenue = ('monthly_charges','sum'),
    user_count = ('customerid','count')
).reset_index()
print(churn_by_subscription)
```

	subscription_type	churn_flag	total_revenue	user_count
0	Organic	0	145.91	9
1	Paid	0	161.95	5
2	Paid	1	12.99	1
3	Refferal	0	13.99	1
4	Refferal	1	60.95	5

In [55]:

```
#5.ARPU- Avg Revenue per user
arpu = df['monthly_charges'].mean()
print('ARPU = ', round(arpu,2))
```

```
ARPU =  18.85
```

calculate customer Age

In [149…

```
# 1. 'dob' column ko datetime format mein badlein
df['dob'] = pd.to_datetime(df['dob'], errors='coerce')

today = pd.Timestamp.now()


df['age'] = (today - df['dob']).dt.days // 365.25

# 4. Age ko clean whole number (integer) mein badlein
df['age'] = df['age'].fillna(0).astype(int)
```

```
# 5. Result check karein
df[['customerid', 'customer_name', 'dob', 'age']].head()
```

Out[149…

	customerid	customer_name	dob	age
0	0002-ORFBO	keshav	1982-04-12	44
1	0003-MKNFE	raghav	1995-11-23	30
2	0004-TLHLJ	lalita	1978-02-15	48
3	0011-IGKFF	mohan	2001-08-30	25
4	0013-EXCHZ	mira	1990-05-05	36

In [57]:
```
# 6. Avg Customer Tenure
# count of days users has used our service : cancellation date else current date
today = pd.Timestamp.today()

df['tenure_days'] = np.where(
        df['cancellation_date'].notna(),

    (df['cancellation_date'] - df['subscription_start_date']).dt.days,

    (today - df['subscription_start_date']).dt.days
)

avg_tenure = df['tenure_days'].mean()
print("Avg Tenure (Days) = ", round(avg_tenure),0)
```

```
Avg Tenure (Days) =  1559 0
```

In [58]:
```
# 7. Revenue at risk - revenue lost from churned users
revenue_at_risk = df.loc[df['churn_flag'] == 1,'monthly_charges'].sum()
print("Revenue at Risk (Rs k = ",revenue_at_risk)
```

```
Revenue at Risk (Rs k =  73.94
```

In [59]:
```
# 8. Esclation rate

escalations_rate = (df['escalations']=='Y').mean()*100
print("Escalation Rate = ",round(escalations_rate,2),"%")
```

```
Escalation Rate =  19.05 %
```

In [60]:
```
# 9 Avg Complaint per user
avg_complaints = df['complaint_count'].sum() / df['customerid'].nunique()
print("Avg Complaints per user = ",round(avg_complaints,2))
```

```
Avg Complaints per user =  0.43
```

In [61]:
```
# 10. Correlation Esclation vs Churn
df['escalations'] = np.where(df['escalations'] == 'Y', 1, 0) # encoding string t
corr_df = df[['escalations', 'churn_flag']].dropna()

correlation = corr_df['escalations'].corr(df['churn_flag'])
print("Correlation between esclation vs churn is = ", round(correlation,2))
```

```
Correlation between esclation vs churn is =  0.77
```

In [62]:
```
# 11. Create a column using existing col - Churn risk
conditions = [
```

```
    (df['churn_score']<50),
    (df['churn_score']>=50) & (df['churn_score']<70),
    (df['churn_score']>= 70)
]

choices = ['Low','med','high']

df['churn_risk']=np.select(conditions,choices,default = 'Unknown')
```

In [63]:
```
df[['churn_risk','churn_score']].tail()
```

Out[63]:

	churn_risk	churn_score
16	med	62
17	Low	27
18	high	99
19	Low	7
20	Low	47

In [64]:
```
df.columns
```

Out[64]:
```
Index(['customerid', 'subscription_start_date', 'subscription_type',
       'renewal_date', 'plan_type', 'contract_type', 'cancellation_date',
       'cancellation_reason', 'monthly_charges', 'cltv', 'churn_score',
       'churn_flag', 'customer_name', 'country', 'state', 'gender', 'dob',
       'complaint_date', 'escalations', 'csat_score', 'complaint_count',
       'tenure_days', 'churn_risk'],
      dtype='str')
```

In [65]:
```
df.head(4)
```

Out[65]:

	customerid	subscription_start_date	subscription_type	renewal_date	plan_type	cont
0	0002-ORFBO	2021-03-15	Refferal	2025-03-15	Standard	
1	0003-MKNFE	2020-08-01	Paid	2024-08-01	Premium	
2	0004-TLHLJ	2022-11-20	Organic	2025-11-20	Basic	
3	0011-IGKFF	2019-05-10	Paid	2025-05-10	Premium	

4 rows × 23 columns

In [ ]:
```

```

4. Visualization using Matplotlib

In [66]:
```
df_visual = df.copy()
```

In [67]:
```
df_visual.shape
```

Out[67]:
```
(21, 23)
```

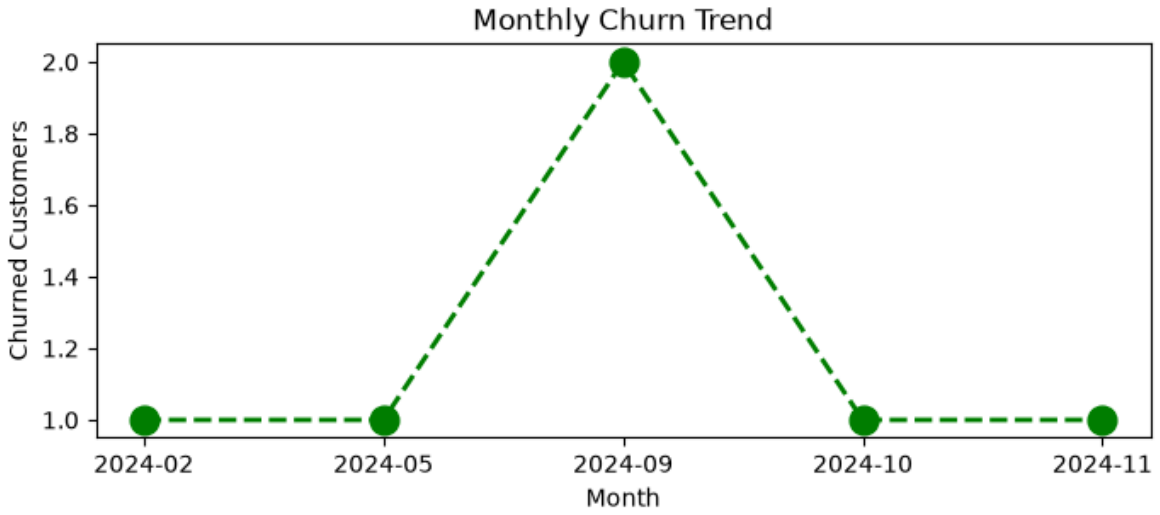
In [68]:
```
df.columns
```

Out[68]:
```
Index(['customerid', 'subscription_start_date', 'subscription_type',
       'renewal_date', 'plan_type', 'contract_type', 'cancellation_date',
       'cancellation_reason', 'monthly_charges', 'cltv', 'churn_score',
       'churn_flag', 'customer_name', 'country', 'state', 'gender', 'dob',
       'complaint_date', 'escalations', 'csat_score', 'complaint_count',
       'tenure_days', 'churn_risk'],
      dtype='str')
```

In [81]:
```
# 4.1 Monthly churn Trend (Time Series KPI)
df_visual['cancellation_month'] =  df_visual['cancellation_date'].dt.to_period('
churn_trend = df_visual[df_visual['churn_flag'] == 1].groupby('cancellation_mont

plt.figure(figsize = (8,3))

plt.plot(churn_trend.index.astype('str'),churn_trend.values, color='green', mark
plt.title('Monthly Churn Trend')
plt.xlabel('Month')
plt.ylabel('Churned Customers')
plt.show()
```

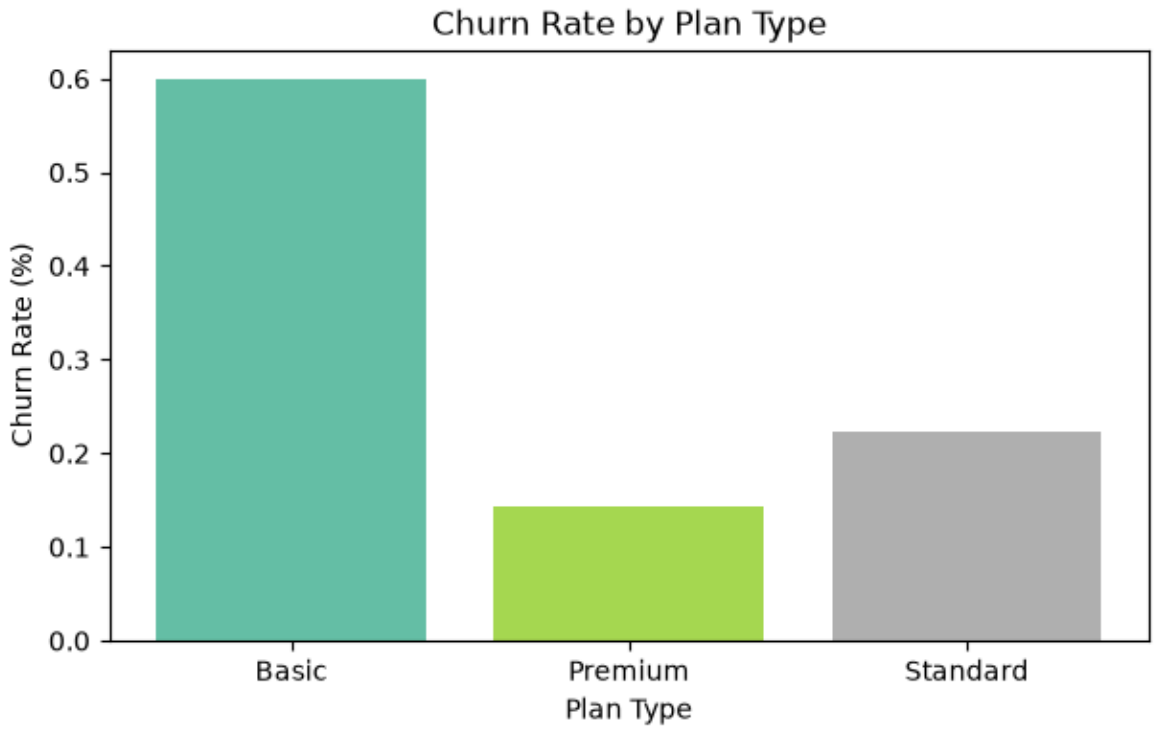


In [89]:
```
# 4.2 Churn by plan type
churn_plan = df_visual.groupby('plan_type')['churn_flag'].mean()

# colors = ['yellow', 'purple', 'blue']
colors = plt.cm.Set2(np.linspace(0, 1, len(churn_plan)))

plt.figure(figsize = (7,4))
plt.bar(churn_plan.index,churn_plan.values ,color = colors)

plt.title('Churn Rate by Plan Type')
plt.xlabel('Plan Type')
plt.ylabel('Churn Rate (%)')
plt.show()
```

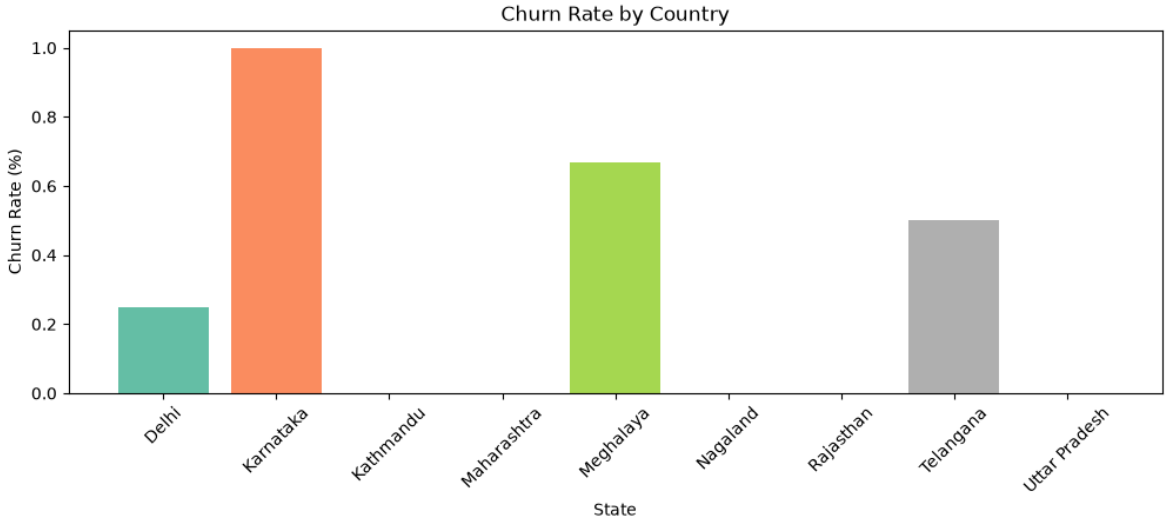


In [90]:
```
# 4.3 Churn by states
churn_plan = df_visual.groupby('state')['churn_flag'].mean()

# colors = ['yellow', 'purple', 'blue']
colors = plt.cm.Set2(np.linspace(0, 1, len(churn_plan)))

plt.figure(figsize=(12,4))
plt.bar(churn_plan.index, churn_plan.values, color = colors)

plt.title('Churn Rate by Country')
plt.xlabel('State')
plt.ylabel('Churn Rate (%)')
plt.xticks(rotation=45)
plt.show()
```



In [ ]:

Visualization using Seaborn

In [112…
```
# Heatmap (Correlation Matrix)
```

In [113…
```
# encoding - convert str to numeric so that we can find corr between features
df_visual.columns
```

Out[113…
```
Index(['customerid', 'subscription_start_date', 'subscription_type',
       'renewal_date', 'plan_type', 'contract_type', 'cancellation_date',
       'cancellation_reason', 'monthly_charges', 'cltv', 'churn_score',
       'churn_flag', 'customer_name', 'country', 'state', 'gender', 'dob',
       'complaint_date', 'escalations', 'csat_score', 'complaint_count',
       'tenure_days', 'churn_risk', 'cancellation_month'],
      dtype='str')
```

In [114…
```
df_visual[['plan_type', 'contract_type', 'churn_score', 'churn_flag', 'churn_ris
```

Out[114…

	plan_type	contract_type	churn_score	churn_flag	churn_risk	escalations
0	Standard	Annual	12	0	Low	0
1	Premium	Annual	91	1	high	1
2	Basic	Monthly	34	0	Low	0
3	Premium	Annual	8	0	Low	0
4	Standard	Monthly	88	1	high	1

In [115…
```
# remove warnings
import warnings
warnings.filterwarnings("ignore")
```

In [116…
```
# incorrect method of encoding - as numbers are not assigned based on priority
df_encoded = df_visual[['plan_type', 'contract_type', 'churn_score', 'churn_flag

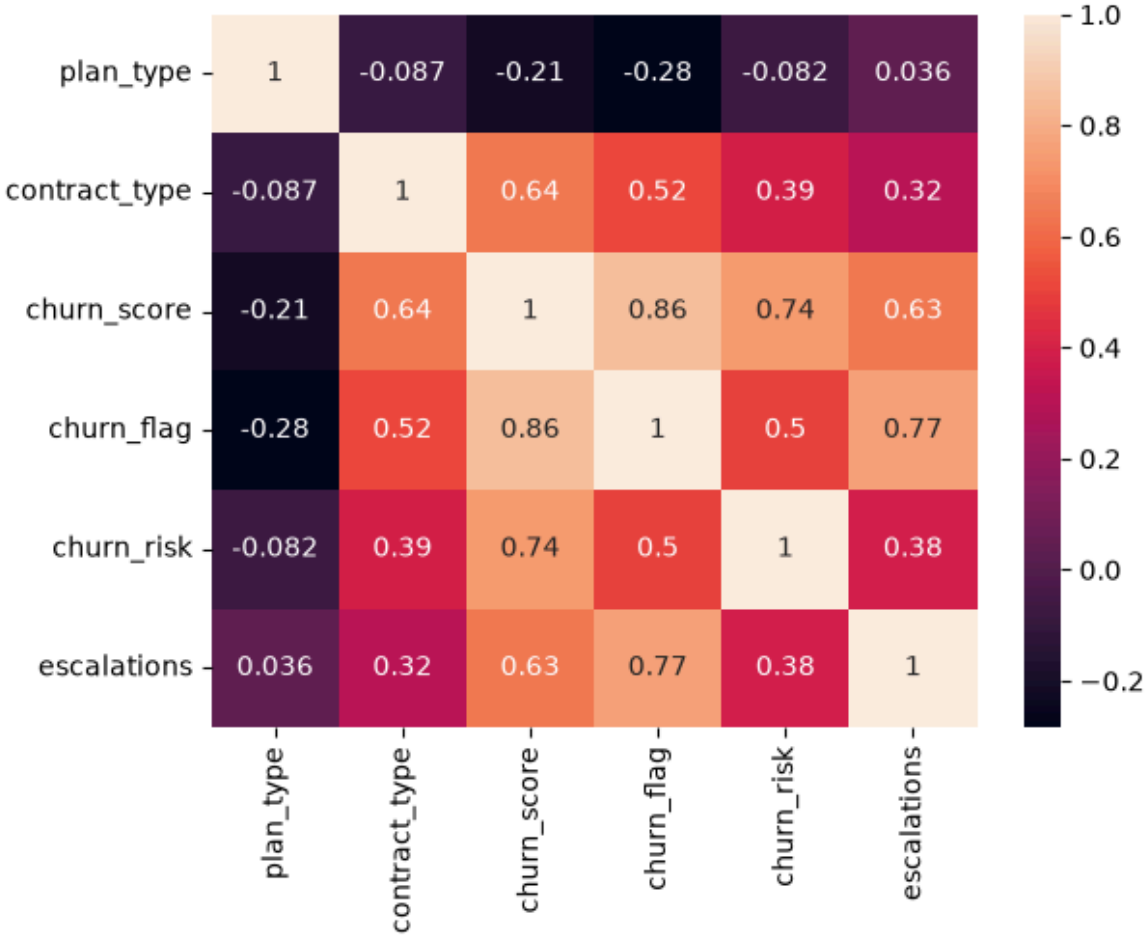
categorial_cols = ['plan_type', 'contract_type', 'churn_risk']

for col in categorial_cols:
    df_encoded[col] = df_encoded[col].astype('category').cat.codes
```

In [117…
```
# Heatmap (correlation matrix)

sns.heatmap(df_encoded.corr(), annot=True)
```

Out[117…
```
<Axes: >
```



In [118…
```
print('plan_type :', df_visual['plan_type'].unique())
print('contract_type :', df_visual['contract_type'].unique())
print('churn_risk :', df_visual['churn_risk'].unique())
```

```
plan_type : <StringArray>
['Standard', 'Premium', 'Basic']
Length: 3, dtype: str
contract_type : <StringArray>
['Annual', 'Monthly']
Length: 2, dtype: str
churn_risk : <StringArray>
['Low', 'high', 'med']
Length: 3, dtype: str
```

In [119…
```
# Correct method of encoding - based on priority
df_encoded = df_visual[['plan_type', 'contract_type', 'churn_score', 'churn_flag

order_mappings = {
    'plan_type' : ['Basic', 'Standard', 'Premium'],
    'contract_type' : ['Monthly', 'Annual'],
    'churn_risk': ['low', 'med', 'high']
    }

for col, order in order_mappings.items():
    df_encoded[col] = pd.Categorical(df_encoded[col].astype('category'), categor
```

In [120…
```
df_visual[['plan_type', 'contract_type', 'churn_score', 'churn_flag', 'churn_ris
```

Out[120…

	plan_type	contract_type	churn_score	churn_flag	churn_risk	escalations
0	Standard	Annual	12	0	Low	0
1	Premium	Annual	91	1	high	1
2	Basic	Monthly	34	0	Low	0
3	Premium	Annual	8	0	Low	0
4	Standard	Monthly	88	1	high	1

In [121…

```
df_encoded.head()
```

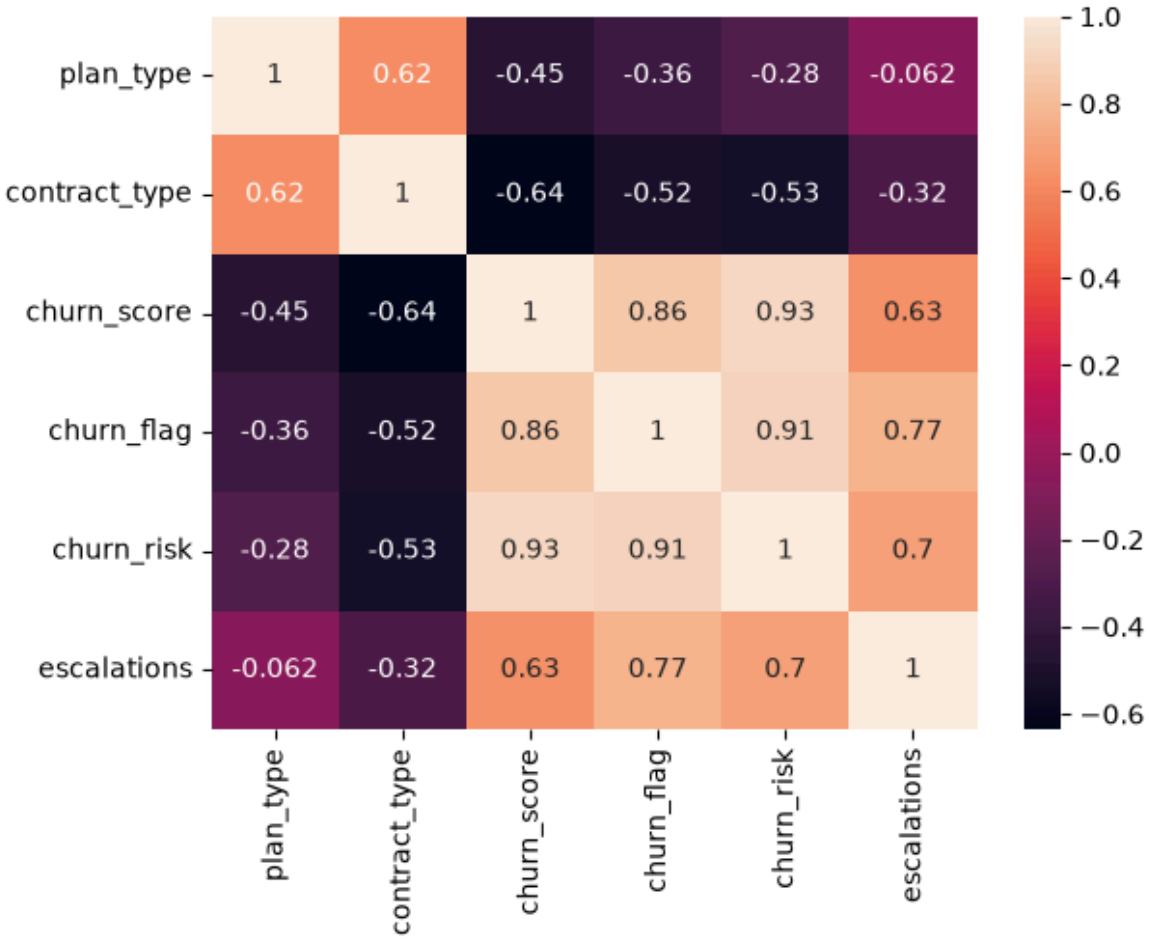
Out[121…

	plan_type	contract_type	churn_score	churn_flag	churn_risk	escalations
0	1	1	12	0	-1	0
1	2	1	91	1	2	1
2	0	0	34	0	-1	0
3	2	1	8	0	-1	0
4	1	0	88	1	2	1

In [122…

```
# Heatmap (correlation matrix)
sns.heatmap(df_encoded.corr(), annot=True)
```

Out[122… <Axes: >



In [123…

```
# Heatmap Using Matplotlib - difficult to plot it

corr_matrix = df_encoded.corr() # Correlation matrix

fig, ax = plt.subplots(figsize=(10, 6)) # Create figure

cax = ax.imshow(corr_matrix, cmap='coolwarm') # Heatmap

fig.colorbar(cax) # Add colorbar

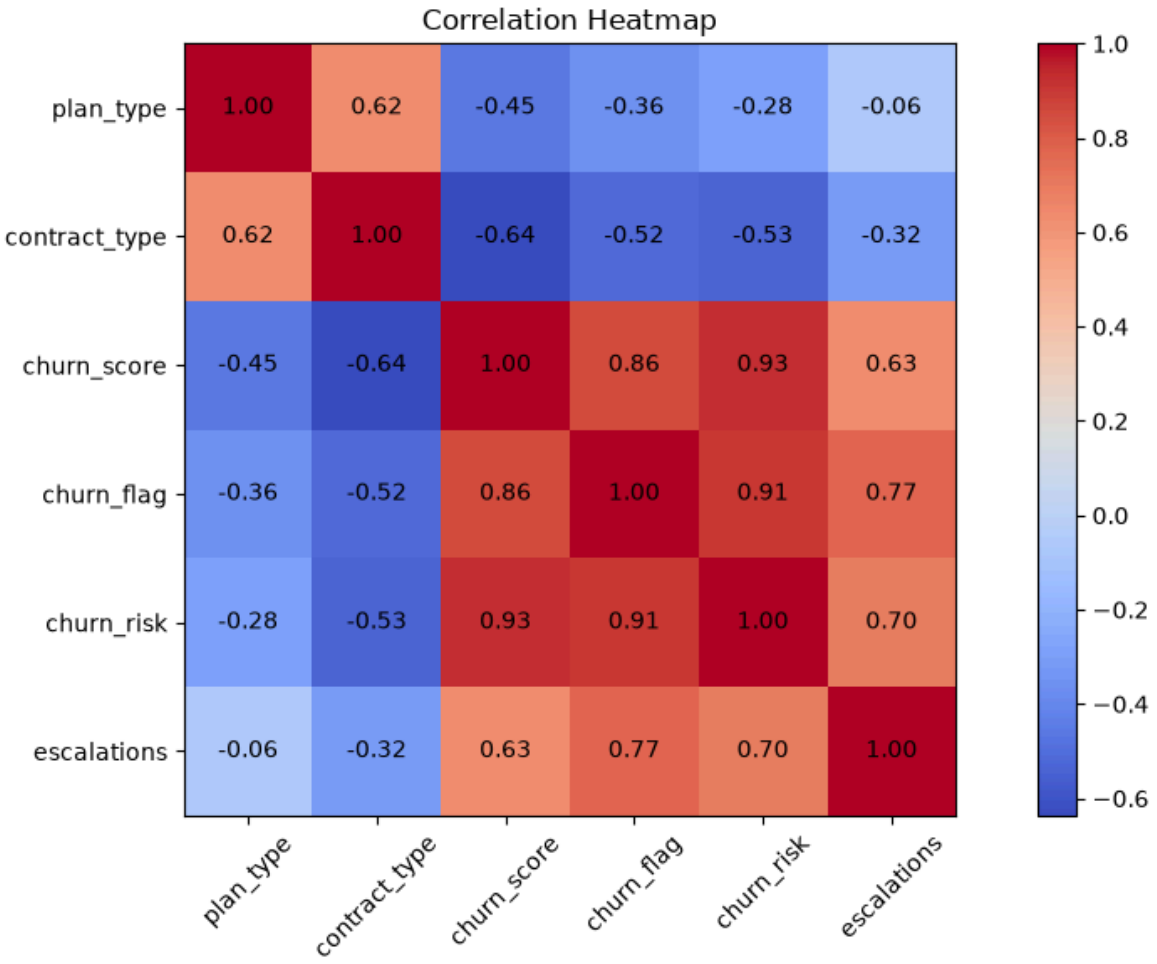
# Axis labels
ax.set_xticks(np.arange(len(corr_matrix.columns)))
ax.set_yticks(np.arange(len(corr_matrix.columns)))

ax.set_xticklabels(corr_matrix.columns, rotation=45)
ax.set_yticklabels(corr_matrix.columns)

# Annotate values inside cells
for i in range(len(corr_matrix.columns)):
    for j in range(len(corr_matrix.columns)):
        ax.text(
            j,
            i,
            f"{corr_matrix.iloc[i, j]:.2f}",
            ha='center',
            va='center'
        )

plt.title('Correlation Heatmap') # Title

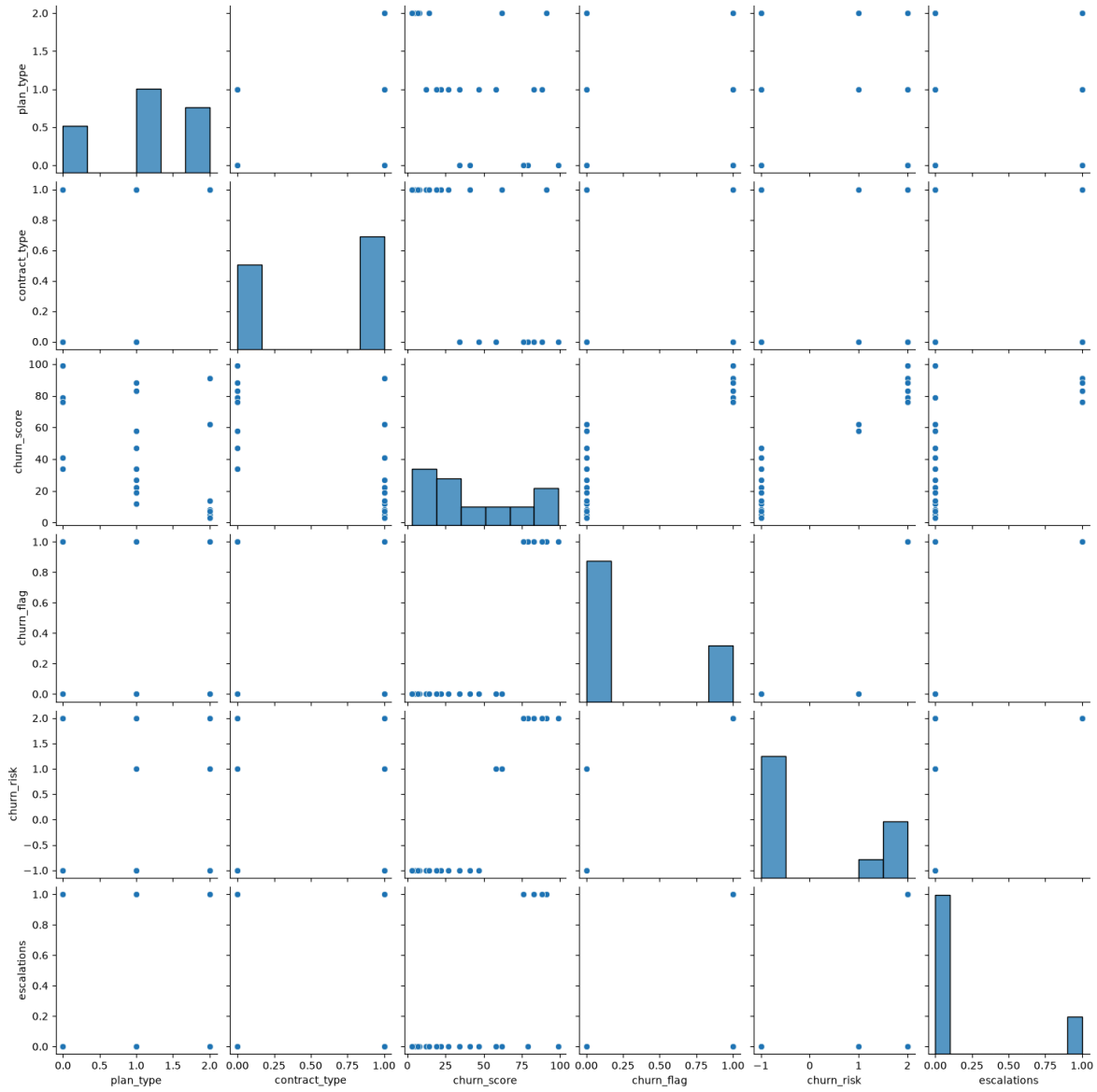
plt.tight_layout()
plt.show()
```



In [124…
```
# pairplot  relationship in dataset

sns.pairplot(df_encoded)
```

Out[124…
```
<seaborn.axisgrid.PairGrid at 0x1d565bf0590>
```



In [125…
```
df_visual.columns
```

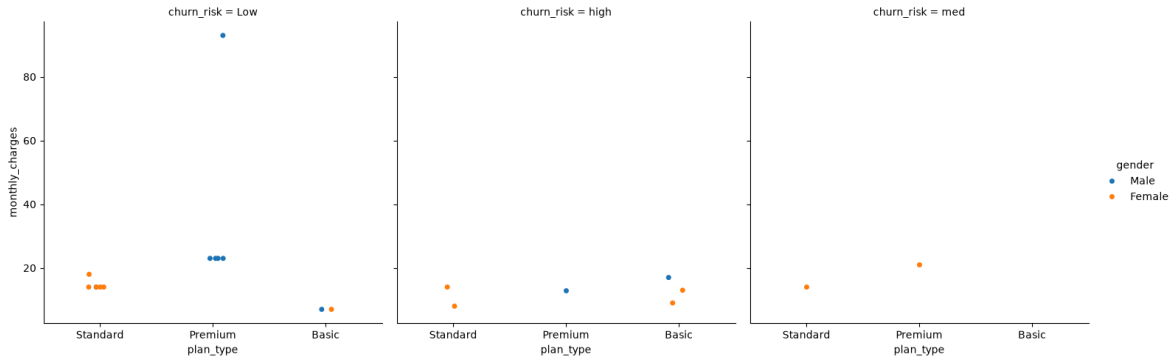
Out[125…
```
Index(['customerid', 'subscription_start_date', 'subscription_type',
       'renewal_date', 'plan_type', 'contract_type', 'cancellation_date',
       'cancellation_reason', 'monthly_charges', 'cltv', 'churn_score',
       'churn_flag', 'customer_name', 'country', 'state', 'gender', 'dob',
       'complaint_date', 'escalations', 'csat_score', 'complaint_count',
       'tenure_days', 'churn_risk', 'cancellation_month'],
      dtype='str')
```

In [126…
```
# catplt/facegrid plot - multi - dimension  comparasion


sns.catplot(data = df_visual,
            x = 'plan_type',
            y = 'monthly_charges',
            hue ='gender',
            col = 'churn_risk',

            )
```

Out[126…
```
<seaborn.axisgrid.FacetGrid at 0x1d568e50440>
```



In [ ]:

pivot Table

In [130…
```
pd.pivot_table(

df_visual,
index = 'plan_type',
values = 'churn_flag',
aggfunc = 'mean'


).reset_index()
```

Out[130…

	plan_type	churn_flag
0	Basic	0.600000
1	Premium	0.142857
2	Standard	0.222222

In [131…
```
df_visual.columns
```

Out[131…
```
Index(['customerid', 'subscription_start_date', 'subscription_type',
       'renewal_date', 'plan_type', 'contract_type', 'cancellation_date',
       'cancellation_reason', 'monthly_charges', 'cltv', 'churn_score',
       'churn_flag', 'customer_name', 'country', 'state', 'gender', 'dob',
       'complaint_date', 'escalations', 'csat_score', 'complaint_count',
       'tenure_days', 'churn_risk', 'cancellation_month'],
      dtype='str')
```

In [132…
```
pd.pivot_table(

df_visual,
index = 'plan_type',
values = ['monthly_charges','customerid','churn_flag'],
aggfunc = {
    'monthly_charges':'sum',
    'customerid':'nunique',
    'churn_flag': 'mean'
}

)
```

Out[132...

plan_type	churn_flag	customerid	monthly_charges
Basic	0.600000	5	52.95
Premium	0.142857	7	218.93
Standard	0.222222	9	123.91

project completed:

In [ ]: